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NETWORK TROUBLESHOOTING AND SECURITY

SECTION-1

1. What is the primary purpose of a firewall in a network security infrastructure?

- a) Encrypting network traffic
- b) Filtering and controlling network traffic
- c) Assigning IP addresses to devices
- d) Authenticating users for network access

2. What type of attack involves flooding a network with excessive traffic to disrupt normal operation?

- a) Denial of Service (DoS)
- b) Phishing
- c) Spoofing
- d) Man-in-the-Middle (MitM)

3. Which encryption protocol is commonly used to secure wireless network communications?

- a) WEP (Wired Equivalent Privacy)
- b) WPA (Wi-Fi Protected Access)
- c) SSL/TLS (Secure Sockets Layer/Transport Layer Security)
- d) AES (Advanced Encryption Standard)

4. What is the purpose of a VPN (Virtual Private Network) in a network security context?

ANS: A VPN is used to create a **secure encrypted connection** over the internet to protect data and provide secure remote access to a network.

SECTION-2

Patch management is the process of regularly updating software and firmware to address security vulnerabilities and improve system performance.

Answer: True

A network administrator should perform regular backups of critical data to prevent data loss in the event of hardware failures, disasters, or security breaches.

Answer: True

Traceroute is a network diagnostic tool used to identify the route and measure the latency of data packets between a source and destination device.

Answer: True

SECTION-3

8. Describe the steps involved in conducting a network vulnerability assessment.

Ans: A network vulnerability assessment is the process of identifying security weaknesses in a network to protect it from cyber attacks and unauthorized access.

Steps Involved in Conducting a Network Vulnerability Assessment

- 1. Identify** **Network** **Assets**
Identify all devices connected to the network such as computers, servers, routers, switches, and firewalls.
- 2. Gather** **Network** **Information**
Collect information about IP addresses, operating systems, applications, and network services.
- 3. Perform** **Network** **Scanning**
Use vulnerability scanning tools to scan the network and detect open ports, weak configurations, and outdated software.
- 4. Identify** **Vulnerabilities**
Analyze the scan results to find security weaknesses that attackers could exploit.

- | | | |
|---|----------------------|-------------------|
| 5. Assess | Risk | Levels |
| Classify vulnerabilities based on their severity and possible impact on the network. | | |
| 6. Apply | Security | Measures |
| Fix vulnerabilities by updating software, applying security patches, changing passwords, and improving firewall settings. | | |
| 7. Verify | and | Re-test |
| Re-scan the network to ensure all vulnerabilities have been properly fixed. | | |
| 8. Prepare | Documentation | and Report |
| Create a report containing identified vulnerabilities, risks, and corrective actions taken. | | |

SECTION-4

9. Demonstrate how to troubleshoot network connectivity issues using the ping command.

The ping command is a network troubleshooting tool used to check whether a device can communicate with another device on the network.

Steps to Troubleshoot Using Ping Command

Step 1: Check Local System

Ping the localhost address:

```
ping 127.0.0.1
```

If replies are received, the TCP/IP protocol is working properly.

Step 2: Check Connection to Gateway/Router

```
ping 192.168.1.1
```

(Example gateway IP)

- If successful → connection to router is working.
- If failed → there may be cable, Wi-Fi, or router issues.

Step 3: Check Another Device in Network

```
ping 192.168.1.10
```

- Helps verify local network communication.

Step 4: Check Internet Connectivity

ping 8.8.8.8

(Google DNS server)

- If successful → internet is working.
- If failed → ISP or internet issue may exist.

Step 5: Check DNS Function

ping www.google.com

- If IP ping works but website ping fails, DNS configuration problem exists.

SECTION-5

10. Discuss the importance of regular network maintenance and the key tasks involved in maintaining network infrastructure.

Regular network maintenance is important to keep the network secure, fast, stable, and reliable. Without maintenance, networks may face problems such as slow speed, security attacks, hardware failure, and downtime. Regular maintenance helps organizations avoid data loss and ensures smooth communication between devices.

Importance of Regular Network Maintenance

1. Improves network performance and speed
2. Detects and fixes problems early
3. Protects the network from cyber attacks and vulnerabilities
4. Reduces downtime and system failures
5. Ensures proper functioning of hardware and software
6. Helps in backup and recovery during emergencies

Key Tasks Involved in Network Maintenance

1. Monitoring network performance and traffic
2. Updating software, firmware, and security patches
3. Checking routers, switches, and cables regularly

4. Performing regular data backups
5. Configuring and updating firewall and antivirus protection
6. Troubleshooting connectivity issues
7. Maintaining documentation of network configurations
8. Conducting security audits and vulnerability checks